

Philipp Kreyenmeier, PhD
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Affiliation:

Computational Neuroimaging and Perception Team, Laboratory of Brain & Cognition
National Institute of Mental Health, National Institutes of Health, Bethesda MD, United States

Education:

- 2018-2024 Graduate Program in Neuroscience, University of British Columbia, Vancouver, Canada
(supervisor: Prof. Dr. Miriam Spering)
PhD in Neuroscience
Dissertation: *Eye movements as a continuous indicator of sensorimotor integration.*
- 2015-2018 Graduate School of Systematic Neurosciences, LMU Munich, Germany
MSc in Neuro-cognitive Psychology
Thesis: *Attention capacity during coordinated eye-hand tasks*
- 2016-2017 University of Glasgow, UK
MSc in Brain Sciences (graduated "with distinction")
Thesis: *Does the winner take all? Processing of hybrid images in early visual cortex: an fMRI study*
- 2012-2015 Ludwig Maximilian University (LMU), Munich, Germany
BSc in Psychology
Thesis: *How visual contexts affect eye and hand movements*

Work Experience:

- 2024-Present **Postdoctoral Visiting Fellow, Laboratory of Brain & Cognition, NIH, Bethesda, MD (Dr. Elisha P. Merriam)**
- 2018 **Graduate student at the Dpt. of Psychology, LMU Munich (Dr. Heiner Deubel)**
Together with Dr. Deubel and senior PhD student Nina M. Hanning, I worked on a project investigating the allocation of attention during preparation of movements using multiple effectors. To this end, we used eye tracking and classical psychophysics.
- 2017 **Visiting research student, Western University, London, Canada (Dr. Brian Corneil)**
From June - September 2017, I worked with Dr. Corneil investigating effects of different spatial frequency targets on stimulus-locked responses in the human upper limb muscles during mid-flight corrections of reaching movements using electromyography (EMG).
- 2016-2017 **Graduate student, University of Glasgow, UK (Dr. Lars Muckli)**
For my master's research project, I worked on a project investigating processing of high and low spatial frequency scene images in early visual cortex using functional magnetic resonance imaging (fMRI) and support vector machine learning (SVM).
- 2015-2016 **Research assistant, German Center for Vertigo and Balance Disorders, Klinikum Grosshadern, LMU Munich (Dr. Paul Taylor)**
In this position I helped conducting studies combining recording of the electroencephalogram (EEG) with transcranial magnetic stimulation (TMS).
- 2015 & 2016 **Visiting research student, UBC, Vancouver, Canada (Dr. Miriam Spering)**
During two research stays (March-June 2015 and February-March 2016) I conducted a study investigating the effects of different visual contexts on smooth pursuit eye movements and manual interception using eye and hand tracking methods.

Fellowships/Awards:

Since 2024	Intramural Research Fellowship, NIH Bethesda, MD, USA
2025	Chan and Peggy Gunn Doctoral Neuroscience Graduate Prize
2018-2023	International Tuition Award, UBC Vancouver
2020-2023	UBC President's Academic Excellence Initiative PhD Award, UBC Vancouver
2022	Margaret L. Adamson Award, Ophthalmology & Visual Sciences, UBC Vancouver
2022	ICICS Travel Fund, ICICS, UBC Vancouver
2022	Graduate Program in Neuroscience Travel Fund, UBC Vancouver
2022	DMCBH General Endowment Award, UBC Vancouver
2022	CVS Symposium Travel Fellowship, Rochester University
2018-2022	International Doctoral Fellowship, UBC Vancouver
2018-2022	International Doctoral Fellowship Tuition Award, UBC Vancouver
2020	Best Presentation Award, Ophthalmology & Visual Science Research Day, UBC Vancouver
2019	Graduate and Postdoctoral Studies Travel Fund, UBC Vancouver
2018-2019	Faculty of Medicine Graduate Award, UBC Vancouver
2018	LMU Forscherpreis (LMU Student Research Award), LMU Munich
2012-2018	German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes)

Peer-reviewed Articles:

- Akande A.A., Kreyenmeier P.*, & Spering M.* (2025). Visual factors that determine sensory uncertainty for the control of rapid manual interceptive movements. *Journal of Vision*, 25(4), 8.
[* shared senior authorship]
- Lapsansky A.B., Kreyenmeier P., Spering M., Wylie D., & Altshuler D.L. (2025). Hummingbirds use compensatory eye movements to stabilize rotary and translatable visual motion. *Proceedings of the Royal Society B*, 292, 20242015.
- Kreyenmeier P., Kumbhani R., Movshon JA, & Spering M. (2024). Shared mechanisms drive ocular following and motion perception. *eNeuro*, 11(6), 1-10.
- Kreyenmeier P., Bhuiyan I., Gian M., Chow H.M., & Spering M. (2024). Smooth pursuit inhibition reveals audiovisual enhancement of fast movement control. *Journal of Vision*, 24(4), 3.
- Kreyenmeier P., Schroeger A., Cañal-Bruland R., Raab M., & Spering M. (2023). Rapid audiovisual integration guides predictive actions. *eNeuro*, 10(8), 1-10.
- Kreyenmeier P., Kämmer L., Fooker, J., & Spering M. (2022). Humans can track but fail to predict accelerating targets. *eNeuro*, 9(5), 1-15. [selected as a feature article]
- Conci M., Kreyenmeier P., Kröll L., Spiech C., & Müller H.J. (2021). The nationality benefit: long-term memory associations enhance visual working memory for color-shape conjunctions. *Psychonomic Bulletin & Review*, 28, 1382-1990.
- Fooker J.*, Kreyenmeier P.*, & Spering M. (2021). The role of eye movements in manual interception: a mini-review. *Vision Research*, 183, 81-90.
[* shared first authorship]
- Kreyenmeier P., Deubel H., & Hanning N.M. (2020). Theory of visual attention (TVA) in action: Assessing premotor attention in simultaneous eye-hand movements. *Cortex*, 133, 133-148.
- Kozak R.A., Kreyenmeier P., Gu C., Johnston K., & Corneil BD (2019). Stimulus-locked responses on human upper limb muscles and online-corrective reaches are preferentially evoked by low spatial frequencies. *eNeuro*, 6(5), 1-17.

Kreyenmeier P., Fookien J., & Spering M. (2017). Context effects on smooth pursuit and manual interception of a disappearing target. *Journal of Neurophysiology*, 118, 404-415.

Other publications:

Kreyenmeier P. & Spering M. (2024). A unifying framework for studying discrete and continuous human movements. *Journal of Neurophysiology*, 131(6), 1112-1114 [Invited commentary]

Kreyenmeier P. (2024). *Eye movements as a continuous indicator of sensorimotor integration*. University of British Columbia. [PhD Thesis].

Conference Proceedings (first-author only):

2026	Poster at the annual meeting of the Vision Sciences Society, St. Pete's Beach, FL, USA
2025	Poster at Gordon Research Conference on Eye Movements, Mount Holyoke, MA, USA
2024	Poster at the Canadian Action & Perception Network Symposium, Edmonton, Canada
2024	Poster at the annual meeting of the Vision Sciences Society, St. Pete's Beach, FL, USA
2023	Talk at Gordon Research Seminar on Eye Movements, Mount Holyoke, MA, USA
2023	Poster at Gordon Research Conference on Eye Movements, Mount Holyoke, MA, USA
2023	Poster at the annual meeting of the Vision Sciences Society, St. Pete's Beach, FL, USA
2022	Poster at Gordon Research Conference on Neurobiology of Cognition, Newry ME, USA
2022	Poster at the Active Vision Symposium, University of Rochester, Rochester, NY, USA
2022	Talk at the annual meeting of the Vision Sciences Society, St. Pete's Beach, FL, USA
2022	Talk at Tagung experimentell arbeitender Psycholog:innen, virtual
2021	Talk at the annual meeting of the Vision Sciences Society, St. Pete's Beach, FL, USA
2019	Poster at Gordon Research Conference on Eye Movements, Lewiston, ME, USA
2019	Poster at the annual meeting of the Vision Sciences Society, St. Pete's Beach, FL, USA
2018	Poster at the meeting of the Neural Control of Movement Society, Santa Fe, NM, USA
2016	Poster at the annual meeting of the Vision Sciences Society, St. Pete's Beach, FL, USA

Invited Talks:

2024	Vision Research Seminar, UBC Vancouver, Canada
2024	Justus Liebig University Giessen, Germany
2022	Friedrich-Schiller University, Jena, Germany
2022	Laboratory of Brain and Cognition Laboratory, NIH, Bethesda, MD, USA
2019	Department of Psychology, UBC Vancouver, Canada

Teaching:

2023	Sessional Instructor COGS 402 (B.A./B.Sc. Cognitive Systems)
2022-2023	Teaching Assistant NRSC 501: Neuroscience II (MSc/PhD Neuroscience)
2022	Teaching Assistant DSCI 561: Linear Regression (MSc Data Sciences)
2022	Instructional Skills Workshop

Leadership & Services:

2023-2025	Co-Chair of the 2025 Gordon Research Seminar on Eye Movements
2023 & 2024	Moderator (Research Trainee Day, Faculty of Medicine, UBC)
2021-2023	Vice-president, Inclusion and Diversity, Vancouver Frontrunners
2022-2023	Member of the DMCBH Awards Committee
2019-2023	Floor Warden Service (ICICS/CS building, UBC)
2020-2022	Graduate Mentor (Faculty of Medicine Student Mentorship Program)
2019-2022	Co-Vice President of Student Mentorship and Wellbeing (Neuroscience Trainee Association, Graduate Program in Neuroscience, UBC Vancouver)
2021-2022	Member of the DMCBH Equity, Diversity, & Inclusion (EDI) Committee

Reviewing:

Journal of Vision (4; 1 "exceptionally good review"); Journal of Neurophysiology (2); Experimental Brain Research (1); Journal of Neuroscience Methods (1); Scientific Reports (2); Psychological Science (1); eLife (1); Perception (1); PLoS One (1); Journal of Experimental Psychology: Human Perception and Performance (1); National Science Foundation (NSF)

Supervision/Mentorship:

Luca Kämmer (summer student; 2020)
Ishmam Bhuiyan (undergraduate student; NSERC-USRA 2020-2021)
Mathew Gian (undergraduate student; 2022-2023)
Abibat Akande (PhD student 2022-2024)
Jessica Chalissery (MSc student 2023-2024)

Summer Schools/Workshops

2025	Cold Spring Harbor Vision Summer School, Cold Spring Harbor, NY, USA
2024-2025	AFNI Bootcamp (Virtual), NIH
2022	Instructional Skills Workshop (ISW), UBC Vancouver
2020	"Neuromatch Academy" (virtual Computational Neuroscience Summer School)
2019	"KINARM Camp", BKin Technology, Kingston ON, Canada
2018	"Visual Search and Selective Attention" symposium, Munich, Germany